

Relativity Without the Lagrangian: Deriving Einstein's and Lorentz's Equations from Orbital Structure and Energy Conservation

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Abstract

We derive the equations of special and general relativity **without variational principles** — no Lagrangian, no Hamiltonian, no action functional. The derivation uses three ingredients only: (1) the **orbit-stabilizer structure** of the admissible transformation group $\mathcal{G}$ acting on space-time; (2) the **generalized Noether theorem** in the form “every kernel symmetry produces a conserved current”; (3) **uniqueness theorems** that constrain the field equations from the conservation and equivariance requirements.

Special Relativity (Theorem 5): The requirement of maximal symmetry ($\kappa = 10$ in 4D) selects constant-curvature spacetimes. Energy conservation under time translation, combined with the mass-shell constraint from the isometry group, yields $E^2 = p^2 c^2 + m^2 c^4$ and the Lorentz transformations.

General Relativity (Theorem 12): We prove that three axioms — (A1) $\text{Diff}(\mathcal{M})$ -equivariance (general covariance), (A2) local energy-momentum conservation $\nabla_\mu T^{\mu\nu} = 0$ (every Killing vector then produces a conserved current), (A3) the field equation is second-order in the metric — uniquely determine Einstein's equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$. In place of the variational Bianchi identity, the derivation imposes local conservation, which forces $\nabla_\mu \mathcal{E}^{\mu\nu} \equiv 0$, and then invokes Lovelock's uniqueness theorem — so no action functional is varied. (General covariance *alone* does not yield the divergence-free condition: $R_{\mu\nu}$ is covariant but not divergence-free.)

The orbital approach reveals that Einstein's equations are not “equations of motion” but **integrability conditions**: they are the unique second-order constraints on the metric that are *compatible* with the existence of conserved Noether

currents for all Killing vectors.

Keywords: orbital derivation, Noether theorem, Bianchi identity, Lovelock theorem, Einstein equations, no variational principle

1 Introduction: The Challenge

Every textbook derives special relativity from Einstein's two postulates (constancy of c + relativity principle) and general relativity from the Einstein-Hilbert action $S = \int R \sqrt{-g} d^4x$. Both derivations involve a variational step: minimizing proper time (SR) or varying the action (GR).

We ask: *can relativity be derived without any variational principle?*

Our answer is yes, using three tools from the twin spaces framework:

1. The **orbit-stabilizer structure** of the symmetry group.
2. The **Noether conservation principle**: kernel symmetry $\Rightarrow$ conserved current.
3. **Uniqueness theorems** (classification of maximally symmetric spaces; Lovelock's theorem).

No action functional appears. No variation is performed. The equations emerge as the *unique constraints compatible with the orbital structure and conservation*.

2 The Orbital Framework

Definition 1 (Spacetime as G -Space). A spacetime is a 4-manifold $\mathcal{M}$ with a Lorentzian metric g and an admissible group $\mathcal{G}$ acting on $\mathcal{M}$. The **kernel** $\text{Ker}(g) = \text{Isom}(\mathcal{M}, g) = \{\psi \in \mathcal{G} : \psi^*g = g\}$ is the isometry group. The **twin space** $\text{Twin}(g) = \mathcal{G}/\text{Ker}(g)$ parameterizes physically distinct deformations.

Definition 2 (Kernel Dimension). The **kernel dimension** $\kappa = \dim(\text{Ker}(g))$ counts the independent symmetries. By Myers–Steenrod: $\kappa \leq n(n+1)/2 = 10$ in 4D.

Definition 3 (Conservation Principle). For each Killing vector $\xi \in \mathfrak{ker}(g)$ (i.e., $\mathcal{L}_\xi g = 0$) and any divergence-free symmetric tensor $T^{\mu\nu}$ ($\nabla_\mu T^{\mu\nu} = 0$), the current $J_\xi^\mu = T^{\mu\nu} \xi_\nu$ is conserved: $\nabla_\mu J_\xi^\mu = 0$.

Remark 4 (No Lagrangian needed). Definition 3 is a *purely geometric* statement: it follows from the Killing equation $\nabla_{(\mu} \xi_{\nu)} = 0$ and the divergence-free condition $\nabla_\mu T^{\mu\nu} = 0$, without any reference to an action functional. The “source” $T^{\mu\nu}$ is assumed to exist as a divergence-free tensor characterizing matter — its existence is a physical axiom, not derived from a Lagrangian.

3 Special Relativity from Maximal Symmetry

Theorem 5 (SR from Orbital Structure). *Assume:*

- (S1) **Maximal symmetry:** $\kappa = 10$ (the space-time has the maximum number of independent isometries).
- (S2) **Energy conservation:** there exists a timelike Killing vector $\xi = \partial_t$ and the associated Noether charge $E = -\int T^{0\nu} \xi_\nu dV$ is conserved.
- (S3) **Homogeneity:** the isometry group acts transitively on spacelike hypersurfaces.

Then:

- (a) $(\mathcal{M}, g)$ has constant sectional curvature K . For $K = 0$: Minkowski space $(\mathbb{R}^{1,3}, \eta)$.
- (b) The isometry group is the Poincaré group $\text{Poinc}(c) = \text{SO}^+(1, 3) \ltimes \mathbb{R}^4$ with a structural constant $c > 0$.
- (c) The energy-momentum of a free particle satisfies:

$$\boxed{E^2 = p^2 c^2 + m^2 c^4} \quad (1)$$

- (d) The coordinate transformations between inertial observers are the Lorentz transformations.

Proof. Step 1: Constant curvature from maximal symmetry. A pseudo-Riemannian manifold with $\kappa = n(n+1)/2$ Killing vectors has constant sectional curvature (classification theorem, cf. Weinberg [3], Ch. 13). In 4D with $\kappa = 10$: $K = 0$ (Minkowski), $K > 0$ (de Sitter), or $K < 0$ (anti-de Sitter). Axiom (S3) and the requirement

of a global timelike Killing vector (S2) select $K = 0$ (Minkowski) in the standard case.

Step 2: Poincaré group from the isometry algebra. For $(\mathbb{R}^{1,3}, \eta)$: the Killing vectors are ∂_μ (4 translations) and $x_\mu \partial_\nu - x_\nu \partial_\mu$ (6 Lorentz generators), totaling 10. They generate $\text{Poinc}(c)$. The parameter c appears as the conversion factor in $\eta = \text{diag}(-c^2, 1, 1, 1)$.

Step 3: Mass-shell from conservation on orbits. A free particle follows a geodesic. Its 4-momentum $p^\mu = m dx^\mu/d\tau$ satisfies $\nabla_p p = 0$ (geodesic equation, which is the *orbit equation* for the particle’s worldline under the Poincaré group — no variational principle needed; it is the condition that the deformation pattern propagates freely, cf. Vol. 17 Theorem 1.5).

The conservation law for the timelike Killing vector $\xi = \partial_t$: $E = -g_{\mu\nu} p^\mu \xi^\nu = c^2 p^0$ (so $E = \gamma m c^2$). For spatial translations ∂_i : $p_i = g_{\mu\nu} p^\mu (\partial_i)^\nu$. The mass-shell condition $g^{\mu\nu} p_\mu p_\nu = -m^2 c^2$ (which is the statement that p lies on the orbit of a massive particle under the Lorentz group) gives:

$$-E^2/c^2 + |\vec{p}|^2 = -m^2 c^2 \quad \Rightarrow \quad E^2 = p^2 c^2 + m^2 c^4 \quad (2)$$

Step 4: Lorentz transformations from orbit structure. The Poincaré group acts on the momentum space p^μ by: $p'^\mu = \Lambda^\mu{}_\nu p^\nu + a^\mu$. The orbits of $\text{SO}^+(1, 3)$ on the mass shell $p^2 = -m^2 c^2$ are the hyperboloids $\{p : g(p, p) = -m^2 c^2, p^0 > 0\}$. The Lorentz boosts Λ_v parameterized by velocity v are:

$$\Lambda_v = \begin{pmatrix} \gamma & -\gamma v/c \\ -\gamma v/c & \gamma \end{pmatrix}, \quad \gamma = (1 - v^2/c^2)^{-1/2} \quad (3)$$

These are determined by the requirement that they preserve $\eta_{\mu\nu} p^\mu p^\nu$ and form a group (orbit-stabilizer structure of $\text{SO}^+(1, 3)$ acting on the mass shell). $\square$

Remark 6 (What was NOT used). No action $S = -mc \int ds$ was minimized. No Lagrangian was varied. The geodesic equation is the **free propagation condition** (deformation shape is parallel-transported), and the mass-shell condition is the **orbit constraint** (the momentum lies on a specific orbit of the Lorentz group). Energy conservation is the Noether charge associated to $\partial_t \in \mathfrak{ker}(\eta)$.

4 General Relativity from Equivariance + Conservation + Uniqueness

We now derive the Einstein equations without any variational principle.

4.1 The Three Axioms

Definition 7 (Axioms for the Gravitational Field Equation). We seek a tensor equation $\mathcal{E}_{\mu\nu}[g] = \kappa T_{\mu\nu}$ relating the metric g to matter T , satisfying:

- (A1) **Diff($\mathcal{M}$)-equivariance**: for all $\psi \in \text{Diff}(\mathcal{M})$, $\mathcal{E}_{\mu\nu}[\psi^*g] = \psi^*\mathcal{E}_{\mu\nu}[g]$. (The field equation has the same form in all coordinate systems — general covariance.)
- (A2) **Local conservation**: the source obeys $\nabla_\mu T^{\mu\nu} = 0$ (equivalence principle). For every Killing vector $\xi \in \mathfrak{ker}(g)$ the current $J_\xi^\mu = T^{\mu\nu}\xi_\nu$ is then conserved, $\nabla_\mu J_\xi^\mu = 0$ (Definition 3).
- (A3) **Second-order**: $\mathcal{E}_{\mu\nu}$ depends on g , ∂g , $\partial^2 g$ only (no higher derivatives).

4.2 The Noether–Bianchi Bridge

The central insight: axioms (A1) and (A2) together force the field equation to satisfy an *identity*, without any variational argument.

Proposition 8 (Divergence-free condition from conservation). *Suppose the field equation $\mathcal{E}_{\mu\nu}[g] = \kappa T_{\mu\nu}$ is required to hold for matter obeying local conservation $\nabla_\mu T^{\mu\nu} = 0$ (A2), across configurations rich enough that g may be arbitrary. Then*

$$\boxed{\nabla_\mu \mathcal{E}^{\mu\nu} \equiv 0 \quad (\text{identically, for all } g).} \quad (4)$$

This is **not** an equation of motion but a constraint on the form of $\mathcal{E}_{\mu\nu}$.

Proof. With $\mathcal{E}_{\mu\nu} = \kappa T_{\mu\nu}$, the conservation law $\nabla_\mu T^{\mu\nu} = 0$ gives $\nabla_\mu \mathcal{E}^{\mu\nu}[g] = 0$ for every metric g sourced by conserved matter. Requiring the theory to be consistent for an arbitrary metric configuration forces the identity to hold for all g . $\square$

Remark 9 (Covariance alone is not enough). It is tempting to claim that Diff($\mathcal{M}$)-equivariance (A1) by itself forces $\nabla_\mu \mathcal{E}^{\mu\nu} \equiv 0$. This is **false**: the Ricci tensor $R_{\mu\nu}$ is a covariant, second-order tensor, yet $\nabla_\mu R^{\mu\nu} = \frac{1}{2}\nabla^\nu R \neq 0$ in general. Covariance constrains the equation to be *tensorial* (the same form in all coordinates); it does not constrain its divergence. In the variational formulation the identity

$\nabla_\mu \mathcal{E}^{\mu\nu} \equiv 0$ is supplied by Noether’s *second* theorem applied to a Diff-invariant *action* [1]. Using no action, we instead obtain the identity from local conservation (Proposition 8): the equivalence principle requires $\nabla_\mu T^{\mu\nu} = 0$, and the field equation must be compatible with it. The contracted Bianchi identity $\nabla_\mu G^{\mu\nu} = 0$ is then the purely geometric statement that $G_{\mu\nu}$ already meets this requirement.

Corollary 10 (Two faces of one requirement). *Given $\mathcal{E}_{\mu\nu} = \kappa T_{\mu\nu}$, the divergence-free condition $\nabla_\mu \mathcal{E}^{\mu\nu} \equiv 0$ and matter conservation $\nabla_\mu T^{\mu\nu} = 0$ are equivalent; under either, the Noether currents $J_\xi^\mu = T^{\mu\nu}\xi_\nu$ are conserved for all Killing vectors ξ (axiom (A2)). Neither follows from covariance alone (Remark 9).*

4.3 Uniqueness: Lovelock’s Theorem

Theorem 11 (Lovelock, 1971 [2]). *In 4 dimensions, the only symmetric, divergence-free ($\nabla_\mu \mathcal{E}^{\mu\nu} = 0$), second-order tensor built from $g_{\mu\nu}$, $\partial_\alpha g_{\mu\nu}$, and $\partial_\alpha \partial_\beta g_{\mu\nu}$ is:*

$$\mathcal{E}_{\mu\nu} = a G_{\mu\nu} + b g_{\mu\nu} \quad (5)$$

where $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$ is the Einstein tensor and a, b are constants.

4.4 The Main Theorem

Theorem 12 (Einstein’s Equations from Orbital Structure). *Axioms (A1)–(A3) uniquely determine the gravitational field equation:*

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}} \quad (6)$$

where $\Lambda = b/a$ is the cosmological constant and $8\pi = \kappa/a$ is fixed by the Newtonian limit.

Proof. Step 1 (A2 $\Rightarrow$ divergence-free): With $\mathcal{E}_{\mu\nu} = \kappa T_{\mu\nu}$, local conservation $\nabla_\mu T^{\mu\nu} = 0$ (A2) forces $\nabla_\mu \mathcal{E}^{\mu\nu} \equiv 0$ (Proposition 8). Covariance (A1) ensures $\mathcal{E}_{\mu\nu}$ is a natural symmetric tensor built from g .

Step 2 ((A1),(A3) $\Rightarrow$ Lovelock): The second-order condition (A3), Diff-covariance (A1), symmetry, and $\nabla_\mu \mathcal{E}^{\mu\nu} = 0$ give, by Lovelock’s theorem: $\mathcal{E}_{\mu\nu} = aG_{\mu\nu} + bg_{\mu\nu}$.

Step 3 (conserved charges): For each $\xi \in \mathfrak{ker}(g)$, $\nabla_\mu T^{\mu\nu} = 0$ and $\mathcal{L}_\xi g = 0$ give $\nabla_\mu (T^{\mu\nu}\xi_\nu) = 0$, so $Q_\xi = \int T^{\mu\nu}\xi_\nu d\Sigma_\mu$ is conserved.

Step 4 (Newtonian limit fixes constants): In the weak-field, slow-motion limit ($g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, $|h| \ll 1$, $v \ll c$): $G_{00} \approx \nabla^2 \Phi / c^2$ where Φ is the

Newtonian potential. Newton's equation $\nabla^2\Phi = 4\pi G\rho$ gives $a = 1$ and $\kappa = 8\pi G/c^4$.

No variational principle was used. The Einstein equations emerge as the *unique* second-order, Diff-equivariant tensor equation whose solutions automatically conserve energy for every Killing vector. $\square$

Remark 13 (The cosmological term and the maximally symmetric vacua). The constant $b = a\Lambda$ in Lovelock's tensor (Theorem 11) is not optional: it is the most general divergence-free, second-order, covariant term of zeroth order in derivatives. Its content unifies this derivation with the special-relativistic one. The maximally symmetric ($\kappa = 10$) vacua ($T_{\mu\nu} = 0$) of $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$ satisfy $R_{\mu\nu} = \Lambda g_{\mu\nu}$ (trace: $R = 4\Lambda$) and constant sectional curvature $K = \Lambda/3$; they are exactly the spacetimes of Theorem 5, Step 1: Minkowski for $\Lambda = 0$ ($K = 0$), de Sitter for $\Lambda > 0$ ($K > 0$), anti-de Sitter for $\Lambda < 0$ ($K < 0$). Thus the $K = 0, > 0, < 0$ trichotomy that selected Minkowski in the SR theorem is precisely the sign of the cosmological term forced by Lovelock in the GR theorem, and special relativity is the $\Lambda = 0$ maximally symmetric sector of general relativity.

5 The Orbital Interpretation

Remark 14 (Einstein's equations as integrability conditions). The standard interpretation: " $G_{\mu\nu} = 8\pi T_{\mu\nu}$ are the equations of motion for the metric." Our interpretation: **Einstein's equations are the integrability conditions for the existence of a consistent family of Noether currents.**

If we demand that:

- Every isometry of g produces a conserved current (A2),
- The field equation is covariant under all diffeomorphisms (A1),
- The equation involves at most second derivatives (A3),

then the Einstein equations are the *only* consistent choice. They are not derived from an action — they are forced by the requirement that the orbital structure (symmetries, orbits, conservation) is self-consistent.

Remark 15 (The role of each axiom).

Axiom	Mathematical content	Physical meaning
(A1)	Diff-equivariance $\Rightarrow \mathcal{E}_{\mu\nu}$ a natural tensor	General covariance (same form in all coords)
(A2)	$\nabla_\mu T^{\mu\nu} = 0 \Rightarrow \nabla_\mu \mathcal{E}^{\mu\nu} = 0$ (independent of (A1))	Local conservation (equivalence principle)
(A3)	Second-order in g $+(A1), (A2) \Rightarrow$ Lovelock	No ghosts, stability Selects $G_{\mu\nu} + \Lambda g_{\mu\nu}$

The three axioms are independent: covariance (A1) does *not* imply the divergence-free condition (A2), as the covariant but non-conserved tensor $R_{\mu\nu}$ shows (Remark 9). All three — covariance, conservation, and second-order — are needed for the Lovelock characterization.

6 SR as the $\kappa = 10$ Limit

Corollary 16 (SR as maximal kernel). *Special relativity is the limit $\kappa \rightarrow 10$ of general relativity: when the kernel is maximal, the curvature vanishes, and the Einstein equations reduce to $T_{\mu\nu} = 0$ (vacuum) on Minkowski space. All 10 Killing vectors produce conserved quantities: energy (1), momentum (3), angular momentum (3), and boost charges (3). The Lorentz transformations are the orbit maps of the Poincaré group on the mass shell.*

Corollary 17 (Kernel dimension as complexity measure). *The kernel dimension κ measures the "simplicity" of the gravitational field:*

- $\kappa = 10$: SR (maximum symmetry, no gravity, 10 conservation laws).
- $\kappa = 4$: Schwarzschild (4 conservation laws: $E, \vec{L}$).
- $\kappa = 2$: Kerr (2 conservation laws: E, L_z).
- $\kappa = 0$: generic GR (no exact conservation, maximum gravitational complexity).

The "amount of gravity" is $10 - \kappa$: the number of broken symmetries, each corresponding to a gravitational degree of freedom.

7 Conclusion

We have derived both special and general relativity without any variational principle:

	Ingredient	Result
SR	Maximal symmetry ($\kappa = 10$) + mass-shell orbit constraint + orbit maps of $\text{SO}^+(1, 3)$	Minkowski space $E^2 = p^2 c^2 + m^2 c^4$ Lorentz transformation
GR	local conservation (A2) + covariance (A1), 2nd-order (A3) + Newtonian limit	$\nabla_\mu \mathcal{E}^{\mu\nu} = 0$ $\mathcal{E} = G_{\mu\nu} + \Lambda g_{\mu\nu}$ (Lovelock) $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi T_{\mu\nu}$

The key insight: Einstein’s equations are not equations of motion but **integrability conditions** — the unique second-order constraints that make the orbital structure (symmetries + conservation) self-consistent. The Lagrangian formulation, while elegant, is *not necessary*: the physics is in the orbits.

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